EE2004: Digital Signal Processing

Tutorial 1

(January 29, 2026)

1. Determine whether the following signals are energy signals, power signals, or neither: (a) $(-0.5)^n u[n]$, (b) $x[n] = u[n]$, (c) $2\,e^{j3n}$, (d) $x[n] = 2^n\,u[n]$. Note that $u[n]$ is the unit step sequence, which is defined as 1 for $n \geq 0$ and 0 for $n < 0$.

2. Let $x[n] = \{-1,\ 0,\ \underset{\downarrow}{1},\ 2,\ 5\}$ (where the arrow indicates the sample corresponding to the index $n = 0$). Sketch and label carefully each of the following signals: (a) $x[4-n]$, (b) $x[n]\,u[2-n]$, (c) $x[n-1]\,\delta[n-3]$, (d) $x[n^2]$, (e) even part of $x[2n+1]$.

3. Let $x[n]$ be a signal such that $x[n] = 0$ for $n \notin \{-2,-1,0,1,2,3,4\}$. For each signal given below determine the values of n for which it is guaranteed to be zero. (a) $x[n+4]$, (b) $x[-n]$, and (c) $x[-n+2]$.

4. Let $x[n] \in \mathbb{R}$. Define $x_e[n] = \dfrac{x[n]+x[-n]}{2}$ and $x_o[n] = \dfrac{x[n]-x[-n]}{2}$. Define $E_x = \sum_n |x[n]|^2$. It is claimed that $E_x = E_{x_e} + E_{x_o}$. Prove or disprove the claim.

5. Sketch the odd and even components of the following signals: (a) $x[n] = 1$ for $0 \leq n < 6$, and (b) $x[n] = \{\ldots, 0, \underset{\downarrow}{0}, 1, 2, 1, 0, \ldots\}$.

6. Find the odd and even components of: (a) $u[n]$, (b) $n\,u[n]$, (c) $\sin\left(\frac{n\pi}{4}\right)$, (d) $\cos\left(\frac{n\pi}{4}\right)$, (e) $\sin\left(\frac{n\pi}{4}\right)\,u[n]$, (f) $\cos\left(\frac{n\pi}{4}\right)\,u[n]$, (g) $\cos\left(\frac{n\pi}{4}+\frac{\pi}{4}\right)$, (h) $u[n+5]-u[n-5]$

7. Let $x[n] \in \mathbb{C}$. The *conjugate even part* of $x[n]$ is defined as

$$x_{ce}[n] = \frac{1}{2}\left(x[n] + x^*[-n]\right)$$

$x_{ce}[n]$ is also called as the *conjugate symmetric part*. Similarly, the *conjugate odd part* of $x[n]$ is defined as

$$x_{co}[n] = \frac{1}{2}\left(x[n] - x^*[-n]\right)$$

$x_{co}[n]$ is also called as the *conjugate anti-symmetric part*. Note that whenever $x[n] \in \mathbb{R}$, $x_{ce}[n] = x_e[n]$ and $x_{co}[n] = x_o[n]$.

Find the conjugate symmetric and conjugate anti-symmetric part of $x[n] = e^{jn}$.

8. Let $x_{ce}[n]$ and $x_{co}[n]$ be the conjugate-even and conjugate-odd parts of $x[n]\,u[n]$. Find the simplified expression of

$$\sum_{n=-\infty}^{\infty} x_{ce}[n] \cdot x_{co}^*[n]$$

9. In the continuous-time framework the complex exponential has the form $x(t) = e^{st}$. Its discrete-time counterpart $x[n]$ can be thought of as the sampled version of $x(t)$, obtained by replacing t by nT, where T is the *sampling period*. Thus, an alternate form for the discrete-time exponential is $x[n] = e^{sTn}$. It is easy to go from one form to the other. Let $x[n] = e^{-n}$. Express it as z_0^n and find z_0. Next, let $x[n] = 2^n$. Express it in the form $e^{c \cdot n}$ and find c.

10. (a) Simplify the expression for $x(t)\delta(t)$ when $x(t)$ equals: (i) t, and (ii) $\sin t$. Hint: For the first case, consider the limit of $t\,\delta_\Delta(t)$ as $\Delta \longrightarrow 0$, where $\delta_\Delta(t)$ is a symmetric pulse of height $1/\Delta$ and width Δ centred at the origin, i.e.,

$$\delta_\Delta(t) = \frac{1}{\Delta}\left[u\left(t + \frac{\Delta}{2}\right) - u\left(t - \frac{\Delta}{2}\right)\right]$$

    (b) What is the value of $\delta(t - \pi)\cos t$?

On a computer, continuous-time signals have to be necessarily represented in sampled form, i.e., replacing t by nT, where T is the sampling interval. A small value of T is required to get good plots and good numerical accuracy; the sampling interval must cover all events of interest. However, too small a value of T will result in a large number of samples, increasing memory and storage requirements.

`MATLAB` and `Scilab` are two very powerful, easy-to-learn, software packages that significantly help in learning concepts taught in this and more advanced courses. Their syntax is very similar. We cannot overstate the importance of learning to use these packages and trying out various computer exercises to fortify the ideas learnt in class. Besides the assigned computer problems, you should explore ideas on your own.

11. **Computer assignment** Given a continuous-time signal specified by

$$x(t) = \begin{cases} 1 - |t| & -1 \le t \le 1 \\ 0 & \text{otherwise} \end{cases}$$

    plot the discrete-time sequence that results from uniform sampling of $x(t)$ for the following sampling intervals: (a) 0.25 s, (b) 0.5 s, and (c) 1 s. That is, $x[n] = x(nT)$, where T is the sampling interval. Plot $x(nT)$ versus t and compare that with the plot of $x[n]$ vs n.

12. **Computer assignment** Consider the DT exponential $\mathrm{Re}\{z^n\}$ obtained by sampling the signal $\mathrm{Re}\{e^{st}\}$ with $T = 1$. Using $z = e^{sT}$, plot $\mathrm{Re}\{z^n\}$, as well as $\mathrm{Re}\{e^{st}\}$ for reference, over $-5 \le n \le 5$ for the cases $s = -0.1$ and $s = -0.1 + j2\pi$. Comment on the results.

13. **Computer assignment** Let $x(t) = \cos(2\pi f_0 t)$. Form $x[n] = x(nT)$, where T is called the "sampling period". Let $f_0 = 50$ Hz. Plot the signal $x[n]$ versus nT for a duration of 40ms for various values of T: 125μs, 1 ms, 2 ms, and 10 ms using the default option (which joins the individual points by straight line); repeat to get "stem" plots. Observe what happens when you plot $x[n]$ versus n in all these cases.

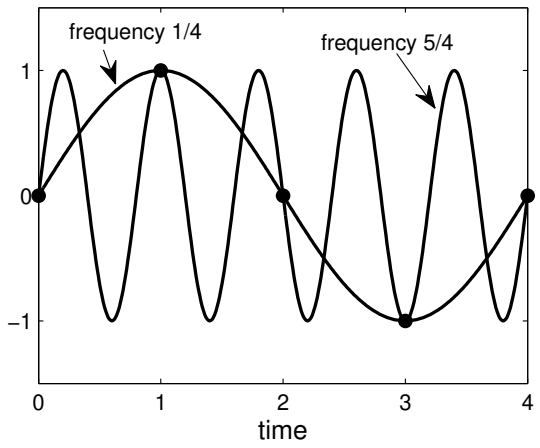
14. **Computer assignment** Plot $(0.9)^n \cos\left(\frac{\pi}{6}n - \frac{\pi}{3}\right)$ and $(1.1)^n \cos\left(\frac{49\pi}{6}n - \frac{\pi}{3}\right)$ for $-10 \le n \le 10$.

15. **Computer assignment** Let $x(t) = A\ \sin(\omega_0 t + \phi)\exp(-at)$. Let $A = 60$, $\omega_0 = 20\pi$, $\phi = 0$, $a = 6$, and $t = [0, 1]$ (in steps of 1 ms). Plot $x(t)$ versus t.

16. **Computer assignment** Let $x(t) = \sin(2\pi f t)$, where f is some given frequency.
    (a) Choose a convenient time interval and a suitable f and plot the signal.
    (b) Plot the signal after time compression by a factor of 2.
    (c) Plot the signal after time expansion by a factor of 2.
    (d) Plot the signal after time reversal.
    (e) Plot the signal after translating it by a non-zero time.

17. **Computer assignment**
    (a) The figure on the right shows two sinusoids with frequencies $f_1 = 1/4$ and $f_2 = 5/4$. They coincide when $t \in \mathbb{Z}$. Reproduce this plot.

    (b) Let $x(t) = \cos(2\pi f t)$ where $f = \pi$ and $T = 1$ (sampling interval). Is the sampled signal periodic? Plot it over the interval $[0, 50]$.



18. **Computer assignment** The continuous-time delta function and its derivatives can be approximated by discrete-time signals by choosing the sampling time small enough. Choose the sampling interval T very small (i.e., 0.01 or less) and the time axis long enough (e.g., $[-1, 1]$) to encompass all events of interest.
    (a) Define a signal u as the unit step and differentiate it numerically to generate a signal d. Plot d and see how it looks.
    (b) Numerically differentiate d twice more to obtain successively signals d_1 and d_2. Plot d_1 and d_2.
    (c) The signal d is an approximation to the delta function δ, and the signals d_1 and d_2 are approximations to $\delta^{(1)}$ and $\delta^{(2)}$ (note: notation-wise, $x^{(n)}(t)$ denotes the n-th derivative; the zeroth derivative is the signal itself). Define the signal ϕ as

$$\phi(t) = \begin{cases} 1 - 4t^2 & -\frac{1}{2} \le t < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

    and check the validity and accuracy of

$$\int_{-\infty}^{\infty} \delta^{(n)}(t)\,\phi(t)\,dt = (-1)^n \phi^{(n)}(0)$$

    for $n = 0$, 1, and 2. Comment on the results.
    (d) Numerically differentiate the signal ϕ twice over the interval $[-1, 1]$. Explain what you find.